

Operational Amplifier(OP–AMP)

EE25Btech11063 (Yalamati Vejith),

EE25Btech11064 (Yojit manral)

March 14, 2026

Abstract

This report details the design and mathematical derivation of an operational amplifier to implement the function $V_{out} = 2V_1 - 3V_2$ using virtual short principle.

1 op–amp

1.1 Aim:

To design an op–amp whose function is $V_{out} = 2V_1 - 3V_2$

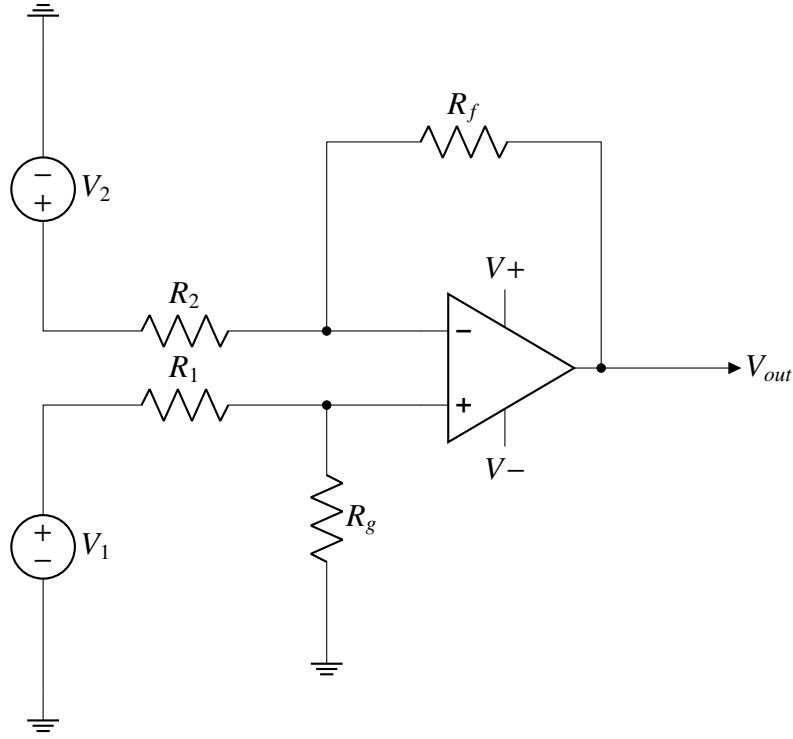
1.2 Materials :

resistors (10K Ω , 30K Ω) , OpAmp (UA741CN), AFG, DSO, voltage supply, breadboard, wires.

1.3 Theory

Below is the difference amplifier circuit configuration used for this operation.

1.3.1 Diagram



Differential Op-amp

1.3.2 Derivation

We derive the output voltage for the general differential amplifier shown in Figure . We assume an ideal op-amp, which gives us two conditions:

1. **Virtual Short:** $V_+ = V_-$ (The voltage at the non-inverting terminal equals the inverting terminal).
2. **Infinite Input Impedance:** $I_+ = I_- = 0$ (No current flows into the op-amp input terminals).

V_1 is applied to a voltage divider formed by R_1 and R_g . Because no current enters V_+ , we apply the voltage divider rule directly:

$$V_+ = V_1 \left(\frac{R_g}{R_1 + R_g} \right) \quad (1)$$

According to the virtual short condition:

$$V_- = V_+ = V_1 \left(\frac{R_g}{R_1 + R_g} \right) \quad (2)$$

Let I_2 be the current through R_2

$$I_2 = \frac{V_2 - V_+}{R_2} \quad (3)$$

The same I_2 flows through R_f according to virtual short principle. Apply KVL to find V_{out}

$$V_{out} = V_+ - R_f I_2 \quad (4)$$

$$V_{out} = V_+ - R_f \left(\frac{V_2 - V_+}{R_2} \right) \quad (5)$$

$$\Rightarrow V_{out} = \left(1 + \frac{R_f}{R_2} \right) V_+ - \left(\frac{R_f}{R_2} \right) V_2 \quad (6)$$

Substitute the value of (1) in (6)

$$V_{out} = \left(1 + \frac{R_f}{R_2} \right) \left(\frac{R_g}{R_1 + R_g} \right) V_1 - \left(\frac{R_f}{R_2} \right) V_2 \quad (7)$$

We want to design the circuit such that:

$$V_{out} = 2V_1 - 3V_2 \quad (8)$$

On comparing the coefficients of V_1 and V_2 in (8) to that of (7) we get

- $R_f = 3R_2$.
- $R_1 = R_g$.

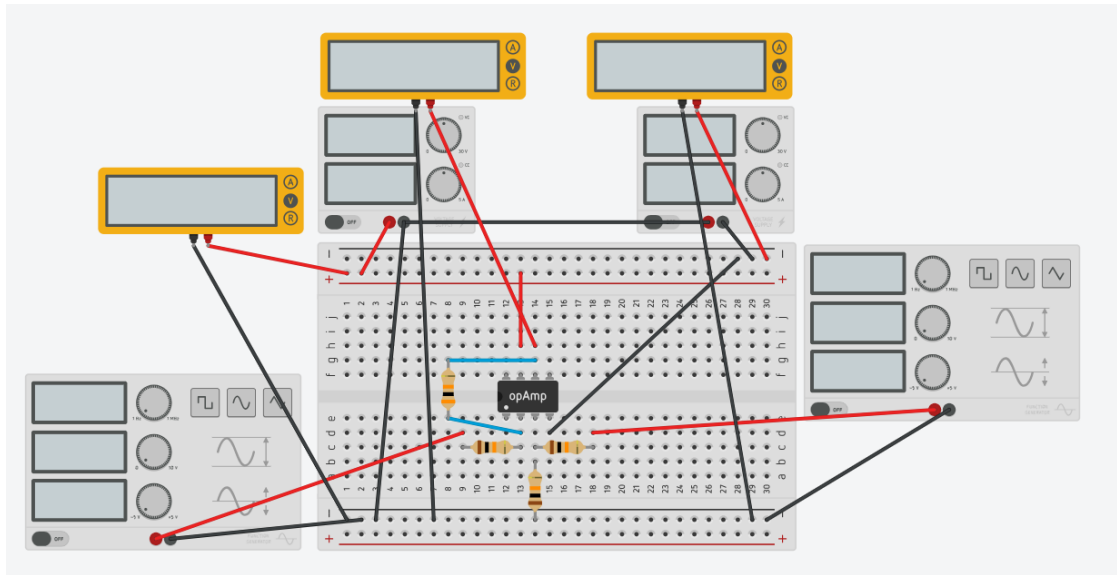
Following the above relations we have taken

- $R_1 = R_g = 10\text{k}\Omega$
- $R_2 = 10\text{k}\Omega \Rightarrow R_f = 30\text{k}\Omega$

Note:

The V_+ and V_- in the above diagram are connected to +15V and -15V of the power supply in the laboratory.

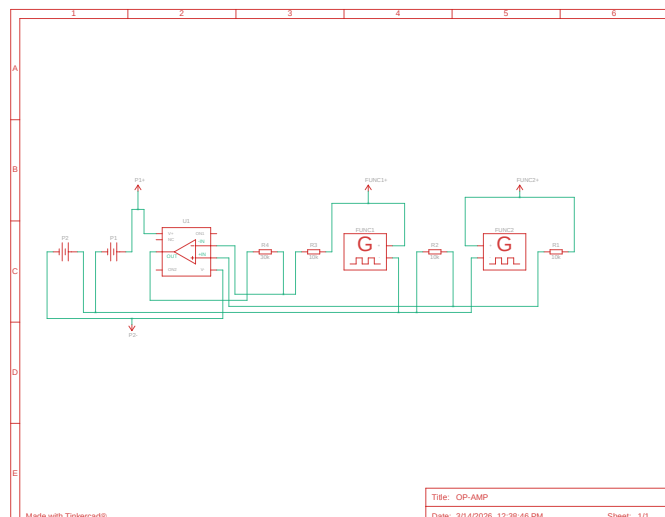
1.4 Circuit Diagram



Circuit diagram

Note:

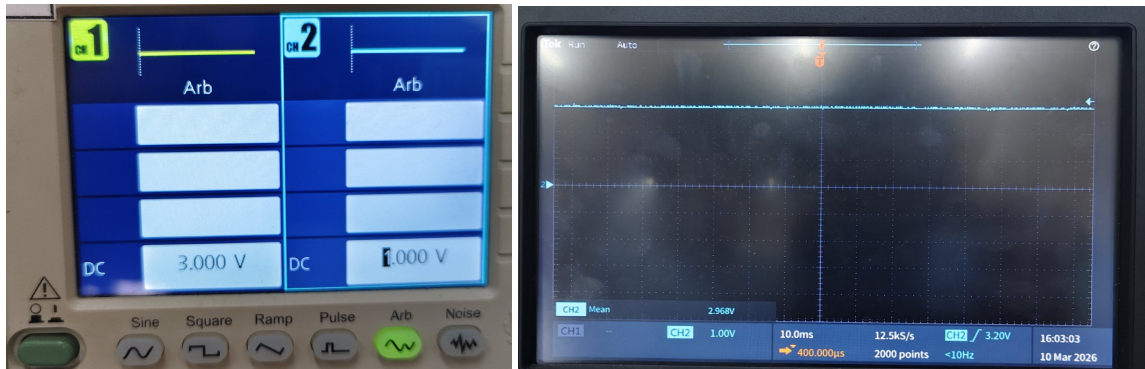
- The multimeters are added in the circuit diagram only to check the value when we do simulations actually multimeter is not used in doing experiment in lab



Schematic view of above circuit diagram

1.5 Observations

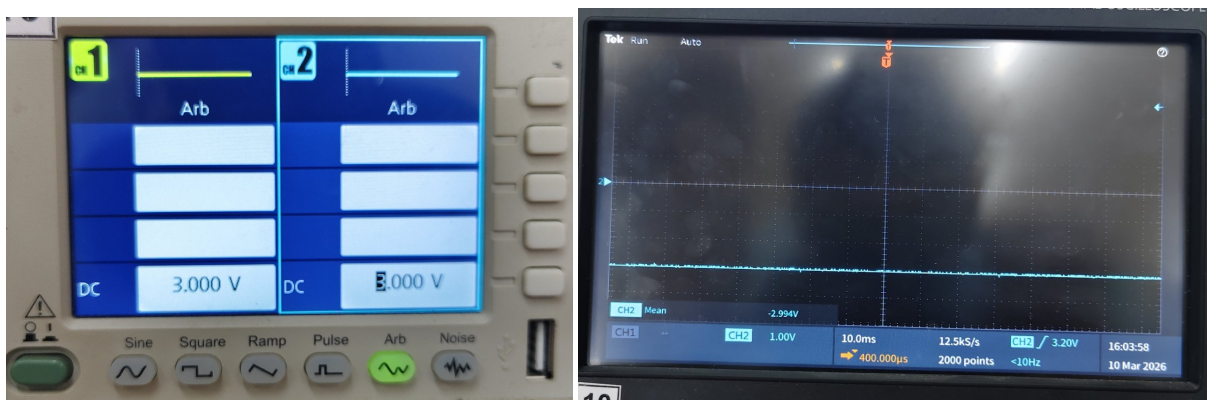
1.5.1 DC input



$$V_1 = 3V \text{ and } V_2 = 1V$$



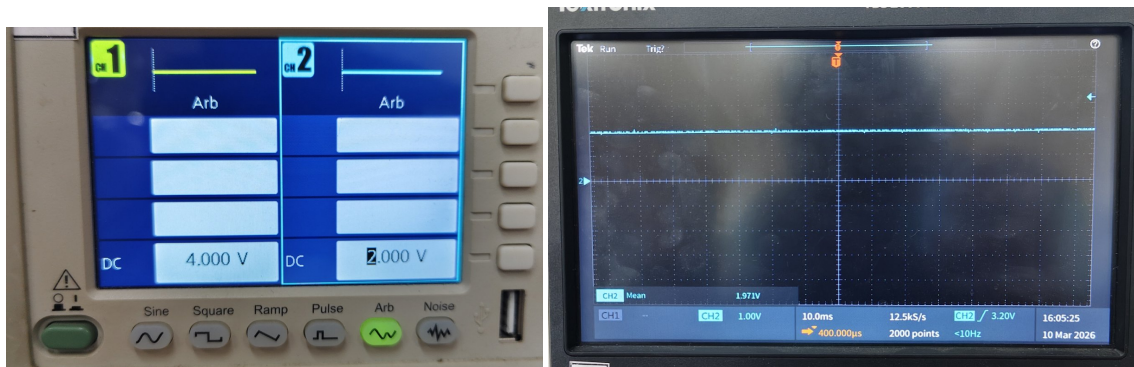
$$V_1 = 3V \text{ and } V_2 = 2V$$



$$V_1 = 3V \text{ and } V_2 = 3V$$



$$V_1 = 4V \text{ and } V_2 = 3V$$



$$V_1 = 4V \text{ and } V_2 = 2V$$

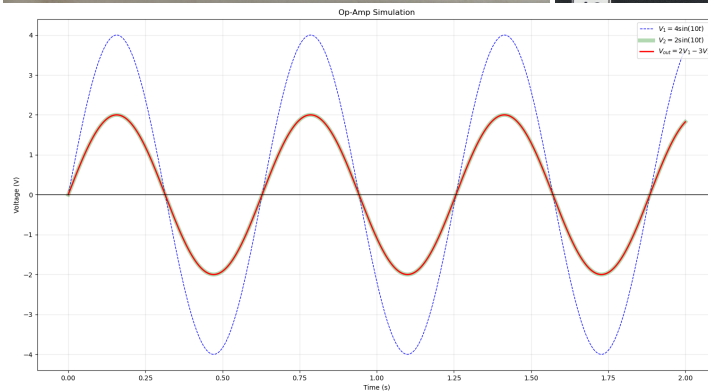
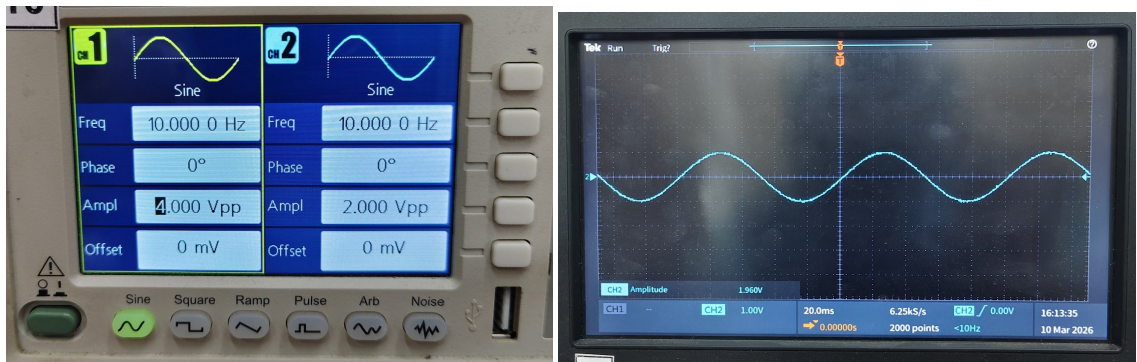
Comparison table

V_1 (V)	V_2 (V)	V_{out} (Exp) (V)	V_{out} (Theo) (V)	Error (%)
3.00	1.00	2.968	3.00	1.067
3.00	2.00	-0.006	0.00	6
3.00	3.00	-2.994	-3.00	0.2
4.00	3.00	-1.001	-1.00	0.1
4.00	2.00	1.971	2.00	1.45

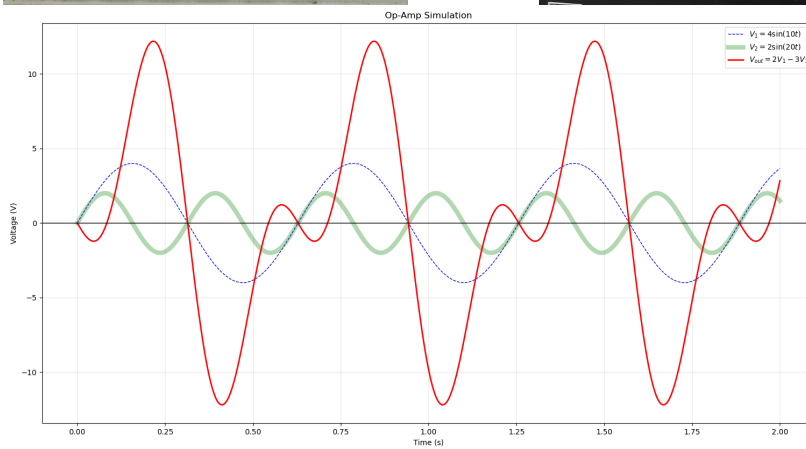
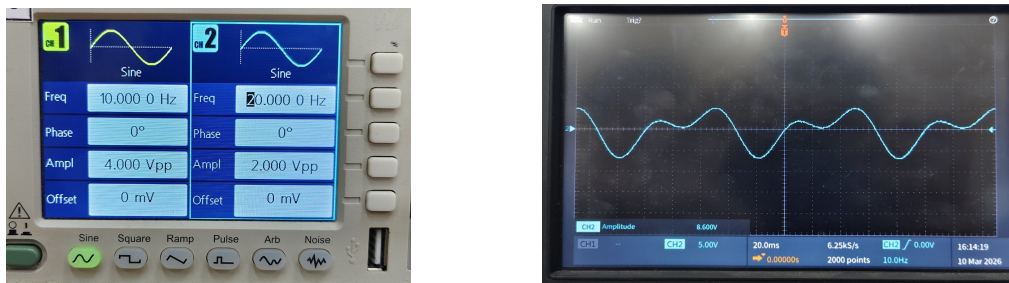
Experimental vs. Theoretical Results for $V_{out} = 2V_1 - 3V_2$

1.5.2 Low frequency sinusoidal inputs

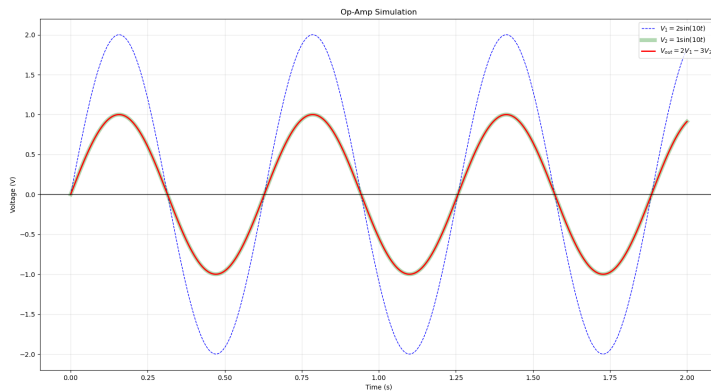
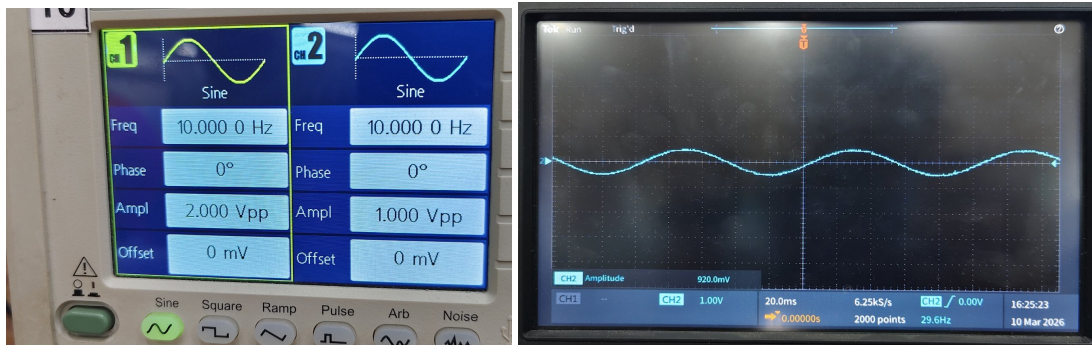
The third photo is the output verified with python plot



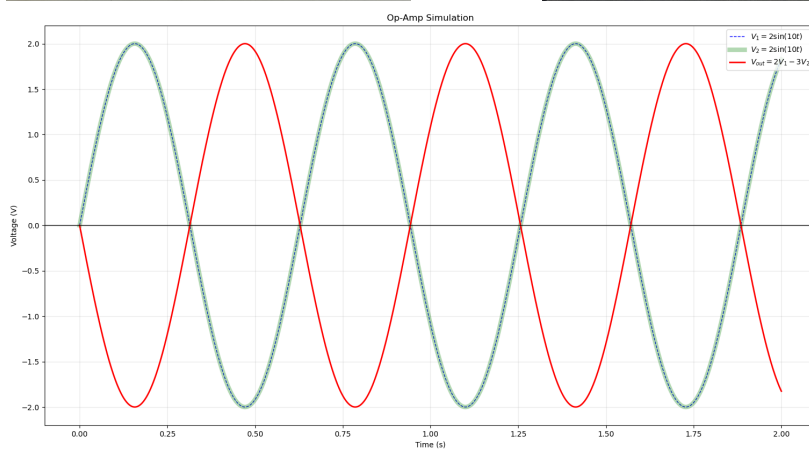
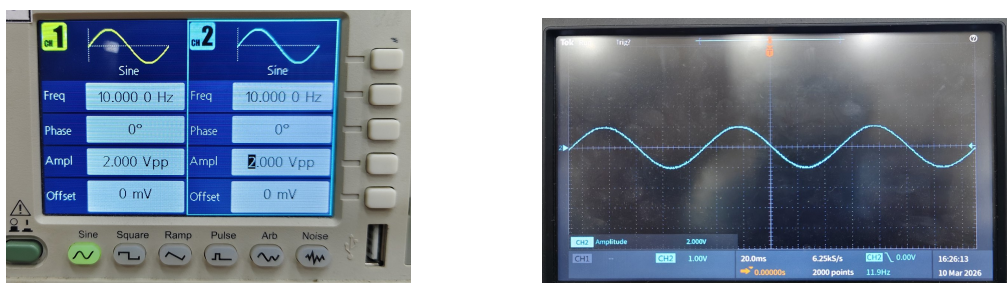
Observation-1 $V_1 = 4 \sin(10t)$ and $V_2 = 2 \sin(10t)$



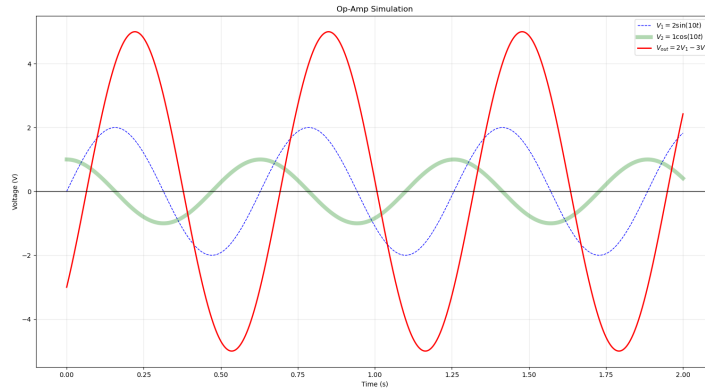
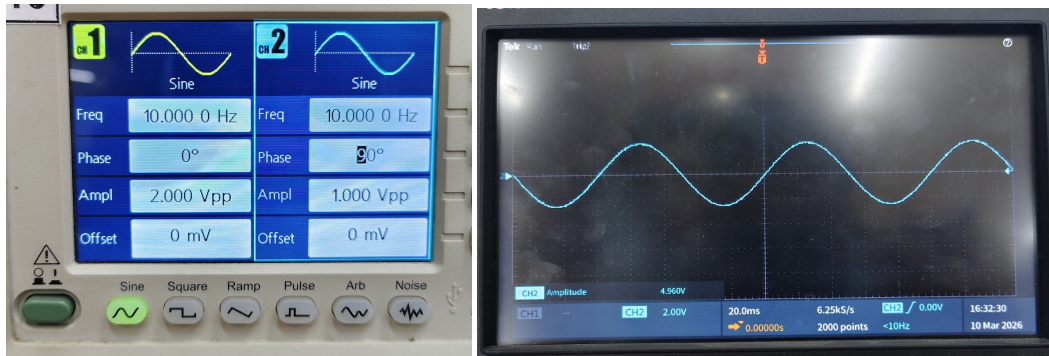
Observation-2 $V_1 = 4 \sin(10t)$ and $V_2 = 2 \sin(20t)$



Observation-3 $V_1 = 2 \sin(10t)$ and $V_2 = 1 \sin(10t)$



Observation-4 $V_1 = 2 \sin(10t)$ and $V_2 = 2 \sin(10t)$



Observation-5 $V_1 = 2 \sin(10t)$ and $V_2 = 3 \cos(10t)$

Comparison table

V_1 (V)	V_2 (V)	Amplitude (Exp) (V)	Phase	Amplitude (Theo) (V)	Error (%)
4.00	2.00	1.968	0°	2.00	1.6
2.00	1.00	0.92	0°	1.00	0.8
2.00	2.00	2.00	0°	2.00	0.0
2.00	1.00	4.96	90°	5.00	0.8

Experimental vs. Theoretical Results for sinusoidal inputs $V_{out} = 2V_1 - 3V_2$

Note:

In the case of observations 1 and 3 V_{out} overlaps exactly with V_2 which we can clearly observe from the python plots of those cases.

1.6 Conclusion

We have successfully designed and op-amp whose function is $V_{out} = 2V_1 - 3V_2$ and successfully tested it for both DC and low frequency sinusoidal inputs. The results obtained experimentally almost matches the theoretical results